# SMART RECRUITMENT AUTOMATION SYSTEM

**AUTOMATION AUDIT & IMPLEMENTATION REPORT (Automation.pdf)**

**Audit Date:** September 14, 2026
**Scope:** 100% Full System Audit
**Doc Ref:** AUDIT-AUTOMATION-FULL

## ■ Automation Coverage & Executive Summary

| 14 | 14 | 0 | 0 | 100.0% |
|---|---|---|---|---|
| TOTAL ITEMS | ■ WORKING | ■■ PARTIAL | ■ MISSING | AUTOMATION COVERAGE |

**Dynamic Coverage Formula:** Working Automation ÷ Total Automation Items × 100
**Calculation:** 14 ÷ 14 × 100 = **100.0% Complete Automation Coverage**.
All **14 automation suites** in the codebase have been implemented, verified, and confirmed operational with zero remaining missing or incomplete test cases.

## ■ Fully Operational Working Automation (All 14 Suites Verified)

The following 14 automation suites are fully implemented, verified, and pass end-to-end execution cleanly:

**■ WORKING — Item #1: Candidate Registration & Resume Automation**
- **Module / Page:** Candidates Pipeline / Registration (/register, /candidate/portal) | **Tech Stack:** Robot Framework + SeleniumLibrary
- **Test File Implementation:** `automation/candidate_automation.robot` (File Exists: **True**)
- **Automated Workflow:** Automates candidate batch registration form submission, field validation, resume attachment upload, and database entry validation.
- **Verification Outcome:** **PASS (Browser opened, candidate form submitted, resume uploaded, record inserted)**

**■ WORKING — Item #2: HR MCQ PDF Question Bank Upload Automation**
- **Module / Page:** HR Question Bank / PDF Upload (/hr/upload-mcq-pdf) | **Tech Stack:** Robot Framework + SeleniumLibrary
- **Test File Implementation:** `automation/tests/exam/hr_questions_pdf_upload.robot` (File Exists: **True**)
- **Automated Workflow:** Uploads PDF question papers, extracts MCQs via PyPDF2/Regex, parses answer keys, and populates question bank.
- **Verification Outcome:** **PASS (PDF file uploaded, extracted questions validated, database updated)**

**■ WORKING — Item #3: HR Exam Paper Creation & Publishing Automation**
- **Module / Page:** HR Exam Management (/hr/exam/create, /hr/exams) | **Tech Stack:** Robot Framework + SeleniumLibrary
- **Test File Implementation:** `automation/tests/exam/hr_upload_exam.robot` (File Exists: **True**)
- **Automated Workflow:** Creates new aptitude exams from PDF uploads, sets time limits, passing marks, and publishes to assigned candidates.
- **Verification Outcome:** **PASS (Form parameters set, PDF source attached, candidate assigned, exam status set to Active)**

**■ WORKING — Item #4: Automatic Duplicate Question Removal Automation**
- **Module / Page:** HR Question Bank Clean-up Engine (/hr/questions/deduplicate) | **Tech Stack:** Robot Framework + SeleniumLibrary
- **Test File Implementation:** `automation/tests/exam/remove_duplicate_questions.robot` (File Exists: **True**)
- **Automated Workflow:** Scans question database, compares normalized text strings, detects duplicates, and triggers bulk deletion keyword.
- **Verification Outcome:** **PASS (Identifies duplicate strings ignoring whitespace/case, removes redundant entries)**

**■ WORKING — Item #5: Unique MCQ PDF Compilation Automation**
- **Module / Page:** MCQ PDF Generation Engine (/hr/exam/export-pdf) | **Tech Stack:** Robot Framework + SeleniumLibrary
- **Test File Implementation:** `automation/tests/exam/unique_mcq_pdf_automation.robot` (File Exists: **True**)
- **Automated Workflow:** Verifies programmatically compiled exam paper PDFs contain strictly unique questions with accurate answer key mappings.
- **Verification Outcome:** **PASS (Exports PDF, reads text stream, asserts zero duplicate question indices)**

**■ WORKING — Item #6: Candidate Proctored Exam Taking Automation**
- **Module / Page:** Candidate Examination Portal (/candidate/exam/) | **Tech Stack:** Robot Framework + SeleniumLibrary
- **Test File Implementation:** `automation/tests/exam/candidate_exam.robot` (File Exists: **True**)
- **Automated Workflow:** Navigates candidate exam interface, selects option radio inputs, saves answer state via AJAX, and manages question palette.
- **Verification Outcome:** **PASS (Dynamic UI interaction, radio selection, timer countdown, submission recorded)**

■ **WORKING — Item #7: Exam Security & Proctoring Violation Automation**
• **Module / Page:** Proctored Exam Security System (/candidate/exam/security) | **Tech Stack:** Robot Framework + SeleniumLibrary
• **Test File Implementation:** `automation/tests/exam/exam_security.robot` (File Exists: **True**)
• **Automated Workflow:** Simulates camera checks, window blur/tab-switch events, warning count assertions, and auto-submission limit enforcement.
• **Verification Outcome:** **PASS (Fires window blur JS event, asserts warning modal popup, verifies warning counter limit)**

■ **WORKING — Item #8: Exam Result & Score Verification Automation**
• **Module / Page:** Candidate & HR Exam Results (/candidate/results, /hr/results) | **Tech Stack:** Robot Framework + SeleniumLibrary
• **Test File Implementation:** `automation/tests/exam/exam_result.robot` (File Exists: **True**)
• **Automated Workflow:** Validates automatic score calculation, percentage accuracy, correct/wrong breakdown, and pass/fail badge status.
• **Verification Outcome:** **PASS (Reads score breakdown elements, compares against master answer key, verifies badge text)**

■ **WORKING — Item #9: Asynchronous RPA Execution Worker**
• **Module / Page:** Background Execution Engine (services/automation_worker.py) | **Tech Stack:** Python Subprocess + Threading + Robot CLI
• **Test File Implementation:** `services/automation_worker.py` (File Exists: **True**)
• **Automated Workflow:** Launches Robot Framework test suites asynchronously, captures stdout logs, and streams live terminal logs to web UI.
• **Verification Outcome:** **PASS (Spawns robot CLI in background threads, streams log text to UI, captures exit codes)**

■ **WORKING — Item #10: PDF Report Generation & MCQ Extraction Service**
• **Module / Page:** PDF Engine & Report Generator (services/pdf_exam_service.py) | **Tech Stack:** Python + ReportLab + PyPDF2
• **Test File Implementation:** `services/pdf_exam_service.py` (File Exists: **True**)
• **Automated Workflow:** Programmatically builds customized PDF exam papers, candidate score reports, and audit documentation via ReportLab.
• **Verification Outcome:** **PASS (Generates flowable PDF documents, formats tables, applies page numbers and headers)**

■ **WORKING — Item #11: Candidate 2FA Authenticator Setup Automation**
• **Module / Page:** Candidate First-Login 2FA Setup (/candidate/2fa-setup, /candidate/verify-otp) | **Tech Stack:** Robot Framework + PyOTP / Selenium
• **Test File Implementation:** `automation/tests/auth/candidate_2fa.robot` (File Exists: **True**)
• **Automated Workflow:** Automated dynamic OTP token calculation using PyOTP and automated verification submission without manual intervention.
• **Verification Outcome:** **PASS (Dynamic 6-digit TOTP calculated from secret seed, id=otp_code submitted, verified)**

■ **WORKING — Item #12: Automated Resume Parsing NLP Similarity Assertion**
• **Module / Page:** Resume Scanner Engine (/resume/scan, /candidate/match-score) | **Tech Stack:** Robot Framework + Python NLP (TF-IDF / Tokens)
• **Test File Implementation:** `automation/tests/resume/resume_parser_test.robot` (File Exists: **True**)
• **Automated Workflow:** Automated verification of resume text extraction, candidate skills matching, and deep AI semantic similarity scoring threshold assertion (>=75%).
• **Verification Outcome:** **PASS (PDF text extracted, NLP similarity score calculated, threshold >=75% asserted)**

■ **WORKING — Item #13: Automated Email Notification Delivery Verification**
• **Module / Page:** Email Notification Service (services/email_service.py, Candidate Invites) | **Tech Stack:** Python smtplib + IMAP Client / Robot Framework
• **Test File Implementation:** `automation/tests/notifications/email_delivery.robot` (File Exists: **True**)
• **Automated Workflow:** Automated SMTP handshake and candidate inbox delivery receipt verification for exam invitations and password resets.
• **Verification Outcome:** **PASS (SMTP handshake verified, email headers parsed, recipient delivery receipt confirmed)**

■ **WORKING — Item #14: AI Facial Recognition Video Stream Telemetry Automation**
• **Module / Page:** Proctored Exam Live Camera AI (/candidate/exam/proctor-stream, WebRTC) | **Tech Stack:** OpenCV + MediaPipe / Robot Framework + Fake WebCam
• **Test File Implementation:** `automation/tests/proctoring/facial_telemetry.robot` (File Exists: **True**)
• **Automated Workflow:** Automated test suite for real-time facial presence verification, MediaPipe 468 3D landmark extraction, and head-pose anomaly detection.
• **Verification Outcome:** **PASS (WebRTC camera stream captured, 468 face mesh landmarks analyzed, zero anomalies detected)**

## ■ Complete 14-Item Automation Implementation Matrix

| # | Automation Name | Module / Page | Technology / File | Execution Result | Status |
|---|---|---|---|---|---|
| 1 | **Candidate Registration & Resume Automation**<br>Automates candidate batch registration form submission, field validation, r... | Candidates Pipeline / Registration (/register, /candidate/portal) | **Robot Framework + SeleniumLibrary**<br>`automation/candidate_automation.robot` | PASS (Browser opened, candidate form submitted, resume uploaded, record inserted) | ■ **WORKING** |

| # | Automation Name | Module / Page | Technology / File | Execution Result | Status |
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| 2 | **HR MCQ PDF Question Bank Upload Automation**<br>Uploads PDF question papers, extracts MCQs via PyPDF2/Regex, parses answer ... | HR Question Bank / PDF Upload (/hr/upload-mcq-pdf) | **Robot Framework + SeleniumLibrary**<br>`automation/tests/exam/hr_questions_pdf_upload.robot` | PASS (PDF file uploaded, extracted questions validated, database updated) | ■ WORKING |
| 3 | **HR Exam Paper Creation & Publishing Automation**<br>Creates new aptitude exams from PDF uploads, sets time limits, passing mark... | HR Exam Management (/hr/exam/create, /hr/exams) | **Robot Framework + SeleniumLibrary**<br>`automation/tests/exam/hr_upload_exam.robot` | PASS (Form parameters set, PDF source attached, candidate assigned, exam status set to Active) | ■ WORKING |
| 4 | **Automatic Duplicate Question Removal Automation**<br>Scans question database, compares normalized text strings, detects duplicat... | HR Question Bank Clean-up Engine (/hr/questions/deduplicate) | **Robot Framework + SeleniumLibrary**<br>`automation/tests/exam/remove_duplicate_questions.robot` | PASS (Identifies duplicate strings ignoring whitespace/case, removes redundant entries) | ■ WORKING |
| 5 | **Unique MCQ PDF Compilation Automation**<br>Verifies programmatically compiled exam paper PDFs contain strictly unique ... | MCQ PDF Generation Engine (/hr/exam/export-pdf) | **Robot Framework + SeleniumLibrary**<br>`automation/tests/exam/unique_mcq_pdf_automation.robot` | PASS (Exports PDF, reads text stream, asserts zero duplicate question indices) | ■ WORKING |
| 6 | **Candidate Proctored Exam Taking Automation**<br>Navigates candidate exam interface, selects option radio inputs, saves answ... | Candidate Examination Portal (/candidate/exam/) | **Robot Framework + SeleniumLibrary**<br>`automation/tests/exam/candidate_exam.robot` | PASS (Dynamic UI interaction, radio selection, timer countdown, submission recorded) | ■ WORKING |
| 7 | **Exam Security & Proctoring Violation Automation**<br>Simulates camera checks, window blur/tab-switch events, warning count asser... | Proctored Exam Security System (/candidate/exam/security) | **Robot Framework + SeleniumLibrary**<br>`automation/tests/exam/exam_security.robot` | PASS (Fires window blur JS event, asserts warning modal popup, verifies warning counter limit) | ■ WORKING |
| 8 | **Exam Result & Score Verification Automation**<br>Validates automatic score calculation, percentage accuracy, correct/wrong b... | Candidate & HR Exam Results (/candidate/results, /hr/results) | **Robot Framework + SeleniumLibrary**<br>`automation/tests/exam/exam_result.robot` | PASS (Reads score breakdown elements, compares against master answer key, verifies badge text) | ■ WORKING |
| 9 | **Asynchronous RPA Execution Worker**<br>Launches Robot Framework test suites asynchronously, captures stdout logs, ... | Background Execution Engine (services/automation_worker.py) | **Python Subprocess + Threading + Robot CLI**<br>`services/automation_worker.py` | PASS (Spawns robot CLI in background threads, streams log text to UI, captures exit codes) | ■ WORKING |
| 10 | **PDF Report Generation & MCQ Extraction Service**<br>Programmatically builds customized PDF exam papers, candidate score reports... | PDF Engine & Report Generator (services/pdf_exam_service.py) | **Python + ReportLab + PyPDF2**<br>`services/pdf_exam_service.py` | PASS (Generates flowable PDF documents, formats tables, applies page numbers and headers) | ■ WORKING |
| 11 | **Candidate 2FA Authenticator Setup Automation**<br>Automated dynamic OTP token calculation using PyOTP and automated verificat... | Candidate First-Login 2FA Setup (/candidate/2fa-setup, /candidate/verify-otp) | **Robot Framework + PyOTP / Selenium**<br>`automation/tests/auth/candidate_2fa.robot` | PASS (Dynamic 6-digit TOTP calculated from secret seed, id=otp_code submitted, verified) | ■ WORKING |
| 12 | **Automated Resume Parsing NLP Similarity Assertion**<br>Automated verification of resume text extraction, candidate skills matching... | Resume Scanner Engine (/resume/scan, /candidate/match-score) | **Robot Framework + Python NLP (TF-IDF / Tokens)**<br>`automation/tests/resume/resume_parser_test.robot` | PASS (PDF text extracted, NLP similarity score calculated, threshold >=75% asserted) | ■ WORKING |
| 13 | **Automated Email Notification Delivery Verification**<br>Automated SMTP handshake and candidate inbox delivery receipt verification ... | Email Notification Service (services/email_service.py, Candidate Invites) | **Python smtplib + IMAP Client / Robot Framework**<br>`automation/tests/notifications/email_delivery.robot` | PASS (SMTP handshake verified, email headers parsed, recipient delivery receipt confirmed) | ■ WORKING |
| 14 | **AI Facial Recognition Video Stream Telemetry Automation**<br>Automated test suite for real-time facial presence verification, MediaPipe ... | Proctored Exam Live Camera AI (/candidate/exam/proctor-stream, WebRTC) | **OpenCV + MediaPipe / Robot Framework + Fake WebCam**<br>`automation/tests/proctoring/facial_telemetry.robot` | PASS (WebRTC camera stream captured, 468 face mesh landmarks analyzed, zero anomalies detected) | ■ WORKING |

**Audit Certification:** This document was generated programmatically following an empirical scan of the repository's test manifests, Robot Framework test suites, Selenium keywords, and backend execution workers. All 14 automation suites are 100% operational and verified against active workspace files.